

BSc Information Technology · Year 3 Project

Project Proposal

# AutoInsight ZA

*Car Market Intelligence Platform*

Data Engineering · Market Analytics · Web Platform · Power BI Visualisation

**Allowing every South African to buy smarter.**

| Student Number | Student Name    | Email Address                | Project Role  |
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March 2026  
Richfield Institute

## Project Snapshot

|               |                                                                |
|---------------|----------------------------------------------------------------|
| Project Name  | AutoInsight ZA: South African Car Market Intelligence Platform |
| Qualification | BSc Information Technology: Year 3 Project                     |
| Core Software | Python (Playwright) · MySQL · HTML/CSS/JS · Power BI           |
| Data Sources  | Cars.co.za · AutoTrader.co.za · MotusSelect.co.za              |
| Target Brands | Top 15 most-listed car brands in South Africa                  |
| Duration      | February - May 2026 (approx. 4 months)                         |

## Problem Statement: Why This Project Matters

**The average South African buying a second-hand car makes un-informed purchasing decisions.**

With no single aggregated platform, buyers must manually browse multiple sites, guess whether a price is fair, and have no way to track how prices have moved over time. Car dealers are aware of this gap and price vehicles to take advantage of this.

South Africa's used-car market processes thousands of transactions per year. Prices are influenced by Rand exchange rate shifts, import duty changes, fuel cost increases, and regional supply-demand shifts: but consumers have access to none of this context at the point of purchase. AutoInsight ZA will be built to change that.

| Without Our Project                                 | With Our Project                                                   | Who Benefits                   |
|-----------------------------------------------------|--------------------------------------------------------------------|--------------------------------|
| No price benchmarks: buyers rely on gut feelings    | Instant fair-value score and pricing data for key makes/model/year | First-time car buyers          |
| Cannot compare prices across multiple sites         | Aggregated data from major SA portals                              | Budget-conscious families      |
| No visibility of price history or trends            | Track price inflation / deflation over time                        | Repeat buyers & upgraders      |
| Provincial price differences are not clear to buyer | Provincial heat maps revealing price variance                      | Out-of-province shoppers       |
| Private vs. dealer pricing gap is not known         | Clear comparison: private seller vs. dealership                    | Value-seekers                  |
| No forecast on future price movements               | ML price predictions per vehicle type and brand                    | Long-term planners & investors |

## Project Structure



|                         |                                                                                                                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer 1: Gathering Data | Python + Playwright scrapes publicly listed vehicle data from Cars.co.za, AutoTrader, and Motus Select across the top 15 SA car brands. Fields captured include: Make, Model, Year, Mileage (KM), Price (ZAR), Province, Body Type, Transmission, Fuel Type, Seller Type, and meta data.                   |
| Layer 2: MySQL Database | A fully normalised MySQL schema will store all scraped data with dimension and fact tables (dim_make, dim_province, fact_listing, fact_listing_price_history, etc.). Creating tables, scrape-run logs, and price-history tracking allows for time-series and trend analysis from the start of the project. |

|                             |                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer 3: Analytics & ML     | Pandas and NumPy for data cleaning and transformation. A scikit-learn price forecasting model (Random Forest Regressor) is trained on listings to create a predicted market value for any vehicle. Model outputs are written back to the database to be shown on the front end.                                                                                            |
| Layer 4: Web Platform       | A lightweight JavaScript front-end site: built with HTML and CSS: will be the consumer-facing interface. Users can search and filter vehicles, view market summaries, and access the Price Estimator tool. The site will be designed to be clean, responsive, and accessible to all South Africans.                                                                        |
| Layer 5: Power BI Dashboard | Power BI's dashboard visualisation is embedded directly into the web platform through an iFrame element, allowing for Power BI's charting functions without re-building from scratch. Dashboard views include: market overview, price trend lines, provincial heat maps, mileage vs. price scatter plots, dealer vs. private-seller comparisons, and bargain-alert panels. |

Technology Stack

|               |                                          |
|---------------|------------------------------------------|
| Scraping      | Python · Playwright · Responses          |
| Database      | MySQL 8.x · SQLAlchemy · MySQL Workbench |
| Processing    | Pandas · NumPy · Scikit-learn            |
| Front End     | Java Script · HTML5 · CSS3               |
| BI / Data Viz | Power BI (iFramed into site)             |
| Dev Tools     | PyCharm · GitHub · python-dotenv         |

High-Level Timeline

|               |                                         |
|---------------|-----------------------------------------|
| Mid Feb       | Scraper development & database schema   |
| Late Feb      | ETL pipeline & data validation          |
| March         | Full rollout & data gathering           |
| April         | Analytics layer & ML price model        |
| March - April | Web platform development                |
| April - May   | Power BI dashboard & iFrame integration |
| May - June    | Testing, documentation & submission     |

Implementation Languages & Frameworks

The project will use a multi-technology stack optimized for each architecture layer:

- Python 3.11+: Data scraping (Playwright, Requests), ETL processing (Pandas, NumPy), ML modelling (Scikit-learn), database interfacing (SQLAlchemy).
- SQL (MySQL 8.x): Relational database management, query optimization, stored procedures, and indexing.
- HTML5/CSS3/JavaScript: Front-end web development without framework dependencies, allowing for lightweight deployment and wide browser compatibility.
- DAX (Power BI): Dashboard measure calculations, data modelling, visualization configuration.

Development tools include PyCharm (Python IDE), MySQL Workbench (database administration), VS Code (web development), GitHub (version control), and python-dotenv (environment configuration management).

## Software Development Lifecycle Model

This project follows Agile development methodology with iterative 2-week sprints, allowing for quick prototyping, continuous feedback from all team members, and refining the project layers.

### Agile Justification

The Agile model is best fit for this project due to:

- Evolving Requirements, in which scraping targets and ML feature engineering benefit from iterative changes and refining based on early-stage data quality checks.
- Incremental Delivery, where each layer (scraping → database → ML → web → BI) can be developed, tested, and validated independently before integration, reducing risk of not meeting project deadlines.
- Team Member Collaboration, where there are regular sprint reviews to allow course correction and priority adjustments aligned with Richfield timelines and due dates.
- Technical Uncertainty, in that the ML model accuracy and web scraping site changes need validation across sprints, making the waterfall's model upfront planning not practical for this project.

### Sprint Structure

| Sprint      | Duration           | Key Deliverables                                                                 |
|-------------|--------------------|----------------------------------------------------------------------------------|
| Sprint 1-2  | Mid-Late Feb       | Scraper development, database schema design, initial data collection             |
| Sprint 3-4  | Early-Mid Mar      | Full brand rollout (15 brands), ETL pipeline, data validation & cleansing        |
| Sprint 5-6  | Late Mar-Early Apr | ML feature engineering, Random Forest model training, prediction API development |
| Sprint 7-8  | Mid-Late Apr       | Web platform (HTML/CSS/JS), search/filter interface, Price Estimator front-end   |
| Sprint 9-10 | Early-Mid May      | Power BI dashboard design, iFrame integration, final testing & documentation     |

### Key Output for Our Project

- Functional web scraper targeting 15+ key South African car brands across 3 major portals (Cars.co.za, AutoTrader, MotusSelect).
- Normalized MySQL database with 100,000+ vehicle listings and historical price-tracking.
- Trained machine learning model achieving high accuracy on the price prediction validation set.
- Consumer-facing web platform (HTML/CSS/JS) with search, filtering, and Price Estimator tool.
- Embedded Power BI dashboard with 4+ analytical views (trends, heat maps, scatter plots, comparisons).
- Comprehensive project documentation: technical report, project presentation, API documentation, GitHub repository.